

1. EPIC

Build and automate CI workflow for Healthcare Appointment Management Platform

2. FEATURES

2.1 Appointment Service API Development

2.1.1 User Story

Ensure service reliability using health check API

Tasks

- Create Flask application
- Implement "/" home endpoint
- Implement "/health" endpoint
- Test health endpoint locally

2.1.2 User Story

Check appointment status to know booking

Tasks

- Implement "/appointment-status" endpoint
 - Return appointment confirmation response
 - Validate API response
 - Test endpoint locally
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2.2 Doctor Availability Tracking

2.2.1 User Story

Doctor availability before booking an appointment

Tasks

- Implement "/doctor-availability" endpoint
 - Return doctor availability response
 - Validate API response
 - Test endpoint locally
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2.3 Automated Testing

2.3.1 User Story

Automated tests to validate API functionality

Tasks

- Install pytest
 - Write test cases for "/" endpoint
 - Write test cases for "/health" endpoint
 - Write test cases for "/appointment-status" endpoint
 - Write test cases for "/doctor-availability" endpoint
 - Run tests locally
 - Fix failing tests
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2.4 Repository Management (GitHub)

2.4.1 User Story

Version control to manage code changes efficiently

Tasks

- Create GitHub repository
 - Add project files (app.py, test_app.py, requirements.txt)
 - Commit code with proper messages
 - Push code to main branch
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2.5 CI Pipeline Setup (GitHub Actions)

2.5.1 User Story

CI pipeline automation for continuous integration

Tasks

- Create .github/workflows directory
 - Add python-app.yml workflow file
 - Configure Python setup in pipeline
 - Install dependencies in pipeline
 - Run pytest in pipeline
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2.6 Build Verification and Validation

2.6.1 User Story

Build pass successfully before deployment

Tasks

- Trigger pipeline on code push

- Monitor GitHub Actions workflow
 - Verify all tests pass
 - Fix pipeline errors if any
 - Confirm successful build execution
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2.7 Application Monitoring

2.7.1 User Story

Monitor application health status

Tasks

- Use "/health" endpoint for monitoring
- Validate service availability
- Ensure correct response is returned